

Electrophilic substitution reactions of benzene (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Describe and explain the mechanism of the reaction between benzene and a charged electrophile, E^+ .
- Describe the features of benzene (C_6H_6) that make it not prone to undergo addition reactions, despite being highly unsaturated.

Section S2.2.12 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/benzene-hl-id-45135/>) covers benzene, which is a compound that differs from other cyclic compounds such as 1,3,5-cyclohexatriene. Both the physical and chemical evidence of benzene are explained, and the reactivity of this compound is discussed later in this section. This might lead you to ask why is benzene not prone to undergoing such addition reactions like 1,3,5-cyclohexatriene even though it has the double bond? The benzene molecule's aromatic (ring-shaped) nature means it is very stable. Understanding the stability of benzene will help you to learn about its unique behaviour and properties, which is an important prerequisite for learning and understanding its physical and chemical evidence. How does stability play a crucial role in determining the type of reaction that benzene will undergo?

Benzene (C_6H_6) has three carbon double bonds and is highly unsaturated. However, this characteristic does not allow it to react like alkenes. Benzene usually undergoes electrophilic substitution reactions in which one or more atoms on the ring are replaced with another atom or group and will conserve the ring of benzene. On the other hand, the instability of benzene due to less common addition reactions will lead to higher energy in products than in the reactants. **Figure 1** shows the electrophilic substitution reaction between the electrophile, E^+ with benzene.

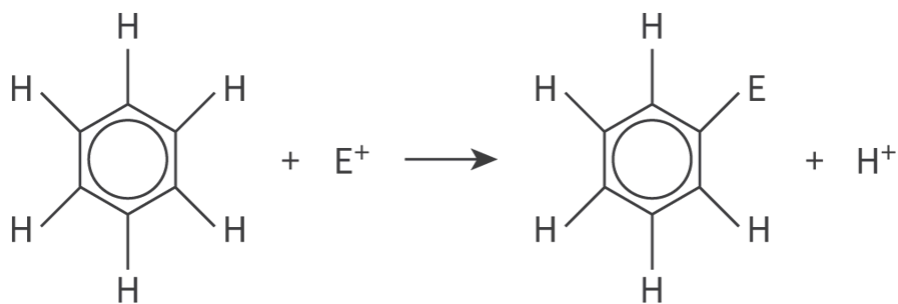


Figure 1. The electrophilic substitution reaction between the electrophile, E^+ with benzene.

 More information for figure 1

The image displays a diagram of an electrophilic substitution reaction. On the left, there is a structural formula of benzene (C_6H_6), represented as a hexagon with alternating double bonds and a circle inside, symbolizing delocalized pi electrons. Each carbon atom in the hexagon is bonded to a hydrogen atom, indicated by a line terminating with an H. To the right of the benzene, there's an arrow pointing towards another benzene structure, which now has one hydrogen replaced by an electrophile group, E. The hydrogen that was replaced is shown detached on the far right as H^+ . The reaction indicates that the electrophile (E^+) attaches to the benzene ring, replacing one of the hydrogen atoms, which becomes a separate hydrogen ion (H^+).

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Electrophilic substitution reactions of benzene involve electrophiles in which benzene plays a role as an electron-rich molecule. Benzene has a ring of six carbon atoms, each of which is bonded to other two carbon atoms and one hydrogen atom. The electrons in the bonds between the two carbon atoms are delocalised around the ring, creating a high electron density region. This leads the ring of benzene being attracted to the electrophile as an electron-deficient species seeking the electrons and forms a bond as a hydrogen departs from the ring structure. All six π electrons of benzene are available to react with an electrophile (E^+) despite their function of maintaining the stability of benzene. Benzene has a stable structure due to delocalisation energy that reduces the repulsion between the electrons. Therefore, the addition reaction is not favoured for benzene as it will interrupt the cloud of delocalised π electrons. This will lead to the instability of the benzene structure. **Figure 2** shows how benzene experiences its loss of symmetry as the positive charge is drawn inside the ring. As the hydrogen atom leaves the carbocation intermediate, the two electrons move back and reproduce the aromatic ring.

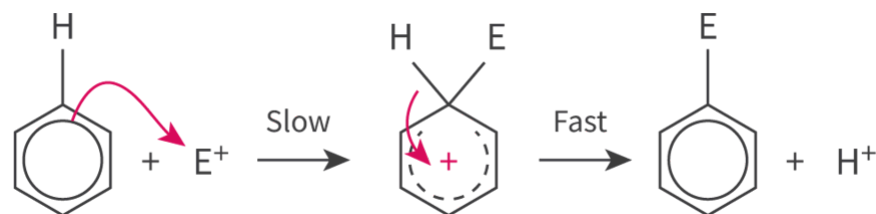


Figure 2. Electrophilic substitution of benzene.

More information for figure 2

The diagram illustrates the electrophilic substitution reaction of benzene. It consists of three main steps:

1. The starting point is a benzene ring with a hydrogen atom (H) attached, alongside an electrophile (E⁺). An arrow indicates the formation of a bond between the benzene and electrophile.
2. The next step shows an intermediate where the benzene ring has bonded with the electrophile (E), but the hydrogen is still present. Within the benzene ring, there is a positive charge signified by "+" inside a circle. An arrow shows movement of electrons inside the ring. This step is indicated as "Slow."
3. The final step shows the benzene ring restored with the electrophile attached, and the hydrogen ion (H⁺) as a separate product. This is labeled as "Fast."

The diagram visually represents how the hydrogen is replaced by the electrophile, illustrating the mechanism of the reaction.

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Think of the sequence of an electrophilic substitution as a table that comfortably seats six people. You, the host (H), and your friends (1 to 5) are seated at the table, comfortable, and having a conversation over a meal. Suddenly, a new elder person, E, appears who wants to join even though there is no more space available. You share your chair with E initially but it is not comfortable, so you give up your chair for E. Now everybody at the table now feels settled and comfortable again to continue enjoying their food and conversation. E is acting as an electrophile that displaces the existence of H as a hydrogen atom. The reaction is completed when the leaving group is displaced by the electrophile.

Study skills

Enhancing specificity of mechanism drawing is crucial in addressing points that frequently negatively affect exam performance. You should note some specific aspects when drawing mechanisms:

1. Curly arrows must have a precise origin and destination: the origin being from the bonding pair or non-bonding pair that are moving and the destination being on the atom they are going to.
2. Transition states and intermediates must show correct bonding arrangements, square brackets and charge (intermediates), or dotted lines showing partially formed/broken bonds (transition states).
3. Make sure you include all final products for the overall reaction equation.

A high activation energy is required to proceed with this reaction and so the process is rather slow (**Figure 3**). The first step occurs when the electrophile is attracted to the electron cloud of the aromatic ring. The unstable carbocation is formed where both the electrophile and the leaving hydrogen are bonded temporarily to the ring. The positive charge is distributed over the ring and the incomplete circle shows that the ring system loses its symmetry. The loss of the proton generates the electrically neutral substitution product with the two electrons from the loss of the carbon–hydrogen bond moving to remake the pi ring system.

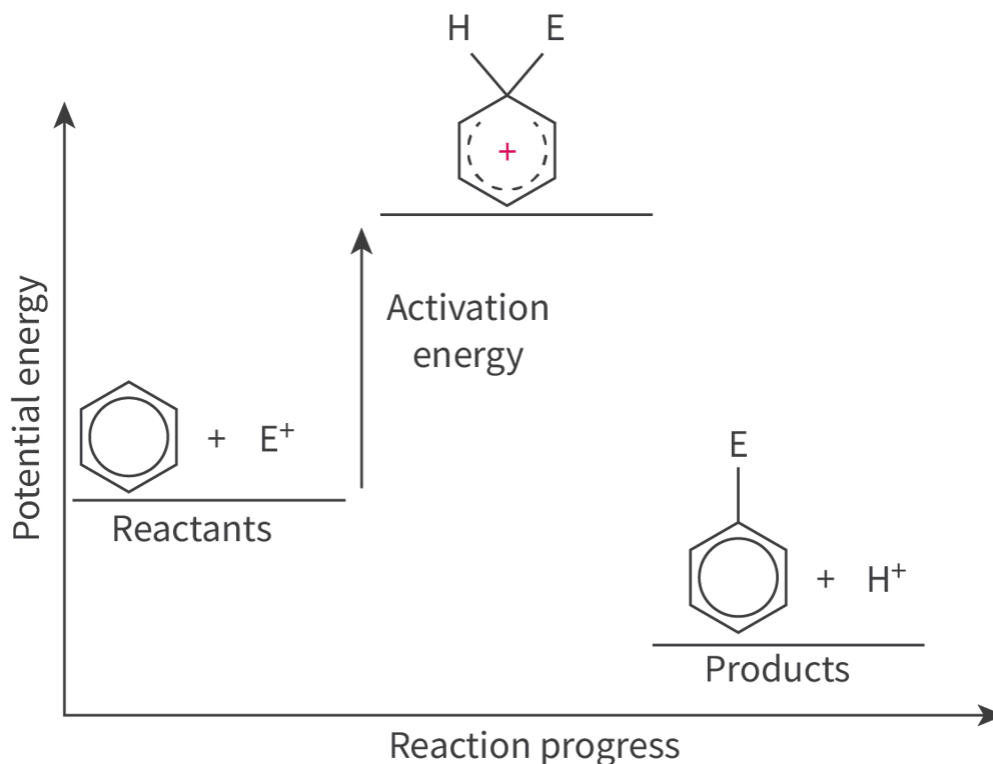


Figure 3. Energy profile of a benzene electrophilic substitution reaction.

 More information for figure 3

The image is a graph representing the energy profile of the benzene electrophilic substitution reaction. The graph has two axes: the Y-axis is labeled "Potential energy," and the X-axis is labeled "Reaction progress." The graph illustrates three major points:

1. **Reactants:** On the left side, labeled as "Reactants," there is a benzene ring with a plus symbol inside, and it is shown combining with an electrophile (E+).
2. **Activation Energy:** The middle section shows the peak of the graph, indicating the transition state. This part is labeled "Activation energy," and it features a benzene ring with double bonds represented as dashed lines. Above this structure, there's an electrophile (E) and a hydrogen atom (H) attached, symbolizing the unstable intermediate carbocation where the positive charge is distributed over the ring.
3. **Products:** On the right side, the graph levels off, indicating the final state after the reaction. Here, the benzene ring is again depicted, now firmly bonded with the electrophile (E) and a hydrogen ion (H+) is released. This section is labeled "Products."

The graph visually represents the energy changes during the reaction process, showing a significant energy barrier due to high activation energy, which corresponds to the endothermic step of forming the unstable intermediate before reaching a lower energy level at the product stage.

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Study skills

The nitration of benzene occurs through a mixture of concentrated sulfuric and nitric acids. Sulfuric acid is used to protonate the hydroxyl group in nitric acid. As a result, the nitrogen-oxygen bond in nitric acid breaks forming a water molecule and the nitronium ion is produced. This then acts as a strong electrophile that readily reacts with benzene.

Here is the chemical equation of the reaction between sulfuric and nitric acid.



Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Pair activity

Your task

Together with your partner, investigate how different electrophiles can add to benzene and the resulting products. Also, compare the stability and reactivity of benzene and its derivatives.

1. Start by reviewing the structure and properties of benzene.
2. Draw the Lewis structure and the resonance structures of benzene.
3. Explain why benzene is more stable than a hypothetical 'cyclohexatriene' with three isolated double bonds.
4. Recalling the general mechanism of electrophilic substitution to benzene, draw the steps involved in the reaction of benzene with bromine in the presence of iron(III) bromide as a catalyst. Label the electrophile, the nucleophile and the intermediate complex. What is the name and structure of the product?
5. Think of other electrophiles that can react with benzene. For each of the following electrophiles, write the equation for the reaction of benzene with them and name the product. Use curly arrows to show the movement of electrons in each step.
 - Chlorine in the presence of iron(III) chloride
 - Nitric acid in the presence of sulfuric acid
 - Sulfur trioxide in the presence of sulfuric acid
 - Ethene in the presence of aluminium chloride.
6. Last, compare and contrast the stability and reactivity of benzene and its derivatives. How does the addition of an electrophile affect the aromaticity of benzene? How does the presence of an electron-withdrawing or electron-donating group on the benzene ring affect its reactivity towards electrophiles? Give examples to illustrate your points.